

INTERNSHIP PROJECT REPORT

Task 2: Library Management System (Python OOP & Automation)

Candidate Name:	KASARABOINA SRINU
Application ID:	KTS020260716223
Internship Track:	Python Development Internship
Duration:	30 Days (20 July 2026 – 19 August 2026)
Supervisor / HR:	Prince Asthana (HR Executive)

1. Executive Summary

The Library Management System is an automated, Object-Oriented Python application engineered to manage book inventories, student/patron memberships, borrowing quotas, loan tracking, and automated overdue fine assessments. The solution provides complete transaction safety, atomic persistence, and rich analytics.

2. Key Architectural Features & Capabilities

- **Catalog Management:** Real-time inventory tracking with physical copy counts and automated stock decrementing.
- **Member Management:** Patron registration with max book borrowing constraints and active issue histories.
- **Automated Loan Lifecycle:** 14-day borrowing cycles with automatic due date computation and return processing.
- **Overdue Fine Engine:** Automatically computes late fees based on exact overdue return timestamps.
- **Search & Filter Engine:** Substring filtering across ISBN, title, author, and category.
- **Analytics:** Metrics on most borrowed books, active loans, and fine revenues.
- **Pytest Coverage:** 100% test pass rate for borrowing, return, and fine calculations.